Raw Guidance

TELEURGENT CARE AI PLATFORM

Simple Developer Explanation

This platform is an asynchronous telemedicine system used in the United States.

Examples of similar platforms:

EasyConsult (Zima Medical)

TeleUrgentCare

K Health

Teladoc asynchronous consults

The system allows patients to request medical care online without phone calls or video calls.

Instead of a live visit, the patient:

1. Enters their symptoms or condition

2. Answers medical questions

3. Uploads photos if needed

4. Submits the consultation

Then a licensed physician reviews the case later.

Typical doctor response time: within 2 hours (but delays may occur).

---

1. HOW HEALTHCARE WORKS IN THE UNITED STATES (IMPORTANT FOR DEVELOPERS)

In the United States:

• Doctors must be licensed in the state where the patient lives
• Patients normally see doctors in person or via telemedicine
• Prescriptions can be sent electronically to pharmacies

However, asynchronous telemedicine allows doctors to review cases without a live visit.

This model works well for simple, common conditions like:

urinary tract infection

pink eye

sinus infection

skin rash

birth control

cold sores

yeast infection

These conditions do not always require a physical exam.

---

2. WHAT THIS PLATFORM DOES

The platform collects structured medical information from the patient.

Then the system sends the case to the doctor in an organized medical summary.

The doctor makes the final decision.

The AI does NOT diagnose the patient.

The AI does NOT give treatment advice to the patient.

Only the doctor provides diagnosis and treatment.

---

3. TWO WAYS A PATIENT CAN START A CONSULTATION

Users can start a consultation in two different ways.

OPTION 1

Condition-Based Consultation

The patient chooses a medical condition they believe they have.

Example:

The patient selects:

• Urinary tract infection
• Acne
• Yeast infection
• Pink eye

The system then:

1. Shows a short description of the condition

2. Shows possible treatments

3. AI asks medical questions related to that condition

Example questions for UTI:

• When did symptoms start?
• Do you have pain when urinating?
• Any fever?
• Any back pain?

• Are you pregnant?

The goal is to collect structured medical history.

OPTION 2

Symptom-Based Consultation

Some patients do not know their diagnosis.

Instead they choose symptoms.

Example:

• cough
• headache
• rash
• nausea
• vaginal discharge

The AI then asks follow-up questions based on the symptoms.

Example:

Patient selects:

Cough

AI asks:

• How long have you had the cough?
• Do you have fever?
• Is the cough dry or with mucus?
• Do you have chest pain?
• Do you smoke?

The goal is again to collect structured medical information.

---

4. IMPORTANT RULE ABOUT AI

The AI must NEVER tell the patient a diagnosis.

The AI must NEVER suggest treatment to the patient.

Instead the AI only:

• asks questions
• collects information
• organizes medical history
• prepares a summary for the doctor

This is very important legally in the United States.

---

5. WHAT THE AI SHOULD DO

The AI acts like a medical intake assistant.

It collects:

Patient demographics

age

sex

location

Medical history

past medical conditions

medications

allergies

surgeries

Current symptoms

when symptoms started

severity

associated symptoms

Risk factors

smoking

alcohol

drug use

travel history

pregnancy

---

6. AI OUTPUT FOR THE DOCTOR

After collecting information, the AI creates a clean medical summary.

Example:

Patient
29-year-old female

Chief Complaint
Painful urination for 2 days

Associated symptoms
• urinary frequency
• urinary urgency
• no fever

• no back pain

Medical history
• no kidney disease
• no recent antibiotics

AI Suggested Differential Diagnosis
• acute cystitis
• urethritis
• vaginitis

Suggested Treatments
• nitrofurantoin
• TMP-SMX
• phenazopyridine

Suggested Tests
• urine culture if symptoms persist

The doctor then:

• reviews the summary
• confirms or changes diagnosis
• approves treatment
• sends prescription if needed

---

7. PHOTOS FOR CERTAIN CONDITIONS

Some medical conditions require photo uploads.

Examples:

• skin rash
• acne
• burns
• eye infections
• cold sores
• shingles

The system should require:

• 2–3 photos
• good lighting
• close-up image
• wider image

The consultation cannot proceed without photos.

---

8. EMERGENCY SAFETY SYSTEM

This platform cannot treat emergencies.

If the AI detects dangerous symptoms, the consultation must stop.

Examples:

• severe chest pain
• stroke symptoms
• severe breathing problems
• loss of consciousness
• suicidal thoughts

The system must immediately show this message:

"Your symptoms may indicate a medical emergency.
Please go to the nearest emergency room or call emergency services immediately."

---

9. PRESCRIPTION LIMITATIONS

The platform cannot prescribe certain drugs.

These include:

• opioids
• controlled substances
• benzodiazepines

• ADHD stimulants

These medications require in-person evaluation.

---

10. WHY THIS SYSTEM WORKS

In U.S. healthcare:

Most telemedicine visits involve simple common conditions.

Examples:

• urinary infections
• sinus infections
• rashes
• yeast infections
• medication refills

These conditions can safely be evaluated using questionnaires.

The doctor reviews the information and makes the decision.

---

11. SYSTEM SIZE REQUIREMENTS

For a fully functional AI platform:

Conditions
120 – 150

Symptoms
200 – 300

AI questions
800 – 1200

This will cover about 95% of urgent care telemedicine visits.

---

12. SIMPLE FLOW OF THE PLATFORM

Step 1
User agrees to Terms and Conditions

Step 2
User selects:

Condition
or
Symptoms

Step 3
AI asks medical questions

Step 4
User uploads photos if needed

Step 5
AI generates medical summary

Step 6
Doctor reviews case

Step 7
Doctor sends:

• diagnosis
• prescription
• recommendations

---

13. IMPORTANT LEGAL RULE

The AI must never act like a doctor.

AI can only:

• collect information
• summarize information

- assist the physician

The doctor always makes the final decision.

Sheets

Link:
TeleUrgentCare_AI_Triage_Import_Pack.xlsx

Link:
Teleurgent_ai_question_bank.xslx

Link:
TeleUrgentCare_Physician_Dashboard_Action_Status_Schema.xlsx

Guide TUC AI

Below is a **developer-ready, database-style master triage rules framework** for your **AI-guided asynchronous TeleUrgentCare / EasyConsult-style platform**.

This version is designed to be:

- **detailed**
- **safe**
- **easy for non-medical developers to build**
- focused on **strong red-flag detection**
- limited to the **symptoms and conditions already listed in your document**

The goal is not for AI to diagnose the patient directly.
The goal is for AI to:

- collect structured history
- stop unsafe consultations
- require photos when needed
- summarize the case for the physician
- generate internal clinical suggestions for the doctor only

---

1. CORE SAFETY PRINCIPLE

The AI must always behave like a **clinical intake assistant**, not like a doctor speaking to the patient.

It must do 5 things well:

1. **Ask relevant questions**
2. **Detect danger early**
3. **Stop consultations when unsafe**
4. **Require photos when necessary**
5. **Create a structured physician-ready summary**

If the system is uncertain whether something is dangerous, it should **escalate rather than continue**.

---

2. MASTER RED-FLAG ENGINE

This is the most important part of the platform.

The AI should run red-flag checks in **3 separate stages**:

Stage A — Before symptom/condition questions

Universal danger screen

Stage B — During question flow

Real-time branching stop logic

Stage C — Before final submission

Final safety scan

If a red flag is triggered in any stage, consultation should stop or be escalated.

2.1 UNIVERSAL RED FLAGS — STOP IMMEDIATELY

If the patient answers yes to any of the following, the consultation must stop immediately:

- severe chest pain
- difficulty breathing
- severe shortness of breath
- stroke-like symptoms
- sudden weakness on one side
- sudden facial droop
- sudden trouble speaking
- loss of consciousness
- fainting with ongoing symptoms
- severe bleeding
- vomiting blood
- coughing up blood
- black stools with weakness or dizziness
- severe abdominal pain
- severe allergic reaction
- swelling of tongue or throat
- suicidal thoughts
- thoughts of harming others
- severe head injury
- sudden paralysis

- seizure
- confusion or altered mental status
- inability to swallow saliva
- drooling with throat symptoms
- severe dehydration with inability to keep fluids down
- fever with stiff neck and severe headache
- vision loss
- severe eye pain
- testicular torsion-like symptoms
- heavy vaginal bleeding with dizziness
- pregnancy with severe abdominal/pelvic pain or heavy bleeding

Emergency message to patient:

Your symptoms may indicate a medical emergency. This platform cannot safely manage this condition. Please go to the nearest emergency room or call emergency services immediately.

2.2 URGENT BUT NOT ALWAYS HARD-STOP

Some cases may still be unsuitable for routine asynchronous review even if not a full emergency.

These should be marked:

- **urgent same-day physician review**
- or **in-person care recommended**

Examples:

- fever in immunocompromised patient
- severe spreading skin infection
- persistent vomiting but still tolerating some fluids
- pelvic pain with fever
- UTI symptoms with flank pain but no vomiting
- severe migraine but no focal deficits
- eye redness in contact lens user
- prolonged cough with weight loss
- new significant memory problem
- major depression without active suicidal ideation
- unilateral leg pain with swelling
- rash with facial swelling but no breathing issues yet

---

3. DATABASE-READY QUESTION ENGINE STRUCTURE

Each question should be stored as a record, not hardcoded in frontend logic.

Recommended fields:

Field	Meaning
question_id	Unique ID
module_type	universal / symptom / condition / modifier / red_flag
parent_topic	fever, cough, UTI, rash, etc.
question_text	Text shown to patient
answer_type	yes_no / single_select / multi_select / text / number / date / photo / severity_scale
answer_options	JSON array if select type
required	true/false
ask_order	Display order
ask_if	Logic condition for showing question
stop_if	Logic condition for hard stop
escalate_if	Logic condition for urgent review
summary_map	Where answer goes in physician summary
negative_map	Tracks key negative findings
photo_required_if	Whether photo upload becomes mandatory
notes_for_devs	Implementation notes

---

4. DECISION ENGINE LOGIC LAYERS

The AI triage engine should use these layers:

Layer 1 — Universal intake

- age
- sex
- state
- allergies
- meds
- pregnancy status
- emergency screen

Layer 2 — Entry mode

- symptom-based
- condition-based

Layer 3 — Primary clinical block

Main complaint logic

Layer 4 — Associated symptom blocks

Only triggered when relevant

Layer 5 — Modifier blocks

- pregnancy
- STI exposure
- chronic illness
- travel
- medication refill
- mental health safety
- photo upload

Layer 6 — Safety block

Final safety scan before submit

---

5. MASTER UNIVERSAL QUESTION TABLE

5.1 UNIVERSAL ELIGIBILITY + SAFETY

ID	Topic	Question	Type	Stop / Escalation Logic	Summary Mapping
U001	eligibility	Are you 18 years or older?	yes_no	stop if no	eligibility_age
U002	eligibility	Are you physically located in a state where our provider is licensed?	yes_no	stop if no	state_eligibility
U003	eligibility	Do you understand this is not emergency care?	yes_no	stop if no	t_and_c
U004	eligibility	Do you agree to asynchronous care without phone or video?	yes_no	stop if no	t_and_c
U005	safety	Are you having severe chest pain right now?	yes_no	stop if yes	red_flag_chest_pain
U006	safety	Are you having trouble breathing right now?	yes_no	stop if yes	red_flag_breathing
U007	safety	Have you fainted or lost consciousness?	yes_no	stop if yes	red_flag_loc
U008	safety	Are you having stroke-like symptoms such as facial droop, weakness, or trouble speaking?	yes_no	stop if yes	red_flag_stroke
U009	safety	Do you have severe bleeding?	yes_no	stop if yes	red_flag_bleeding
U010	safety	Do you have severe abdominal pain?	yes_no	stop if yes	red_flag_abdominal

U011	safety	Are you having a severe allergic reaction or swelling of your face, lips, tongue, or throat?	yes_no	stop if yes	red_flag_allergy
U012	safety	Are you having thoughts of harming yourself or others?	yes_no	stop if yes	red_flag_psych
U013	safety	Have you had a severe head injury?	yes_no	stop if yes	red_flag_head_injury
U014	safety	Do you have sudden paralysis or sudden inability to move part of your body?	yes_no	stop if yes	red_flag_paralysis

---

5.2 UNIVERSAL HISTORY

ID	Topic	Question	Type	Escalation Logic	Summary Mapping
U020	demographics	What is your age?	number	escalate if elderly with high-risk symptoms	age
U021	demographics	What sex were you assigned at birth?	select	none	sex
U022	demographics	Which state are you in right now?	select	stop if unsupported	state
U023	history	Do you have any chronic medical conditions?	multi_select/text	escalate if immunocompromised + infection symptoms	pmh
U024	history	Have you had any surgeries?	text	none	surgeries

U025	history	What medications are you currently taking?	text	interaction checks later	meds
U026	history	Do you have any medication allergies?	text	prescribing alert later	allergies
U027	pregnancy	Could you be pregnant?	yes_no/not_applicable	escalate for bleeding/pelvic pain/vomiting	pregnancy_status
U028	social	Do you smoke?	yes_no/former	affects respiratory assessment	smoking
U029	social	Do you drink alcohol?	yes_no	affects GI/neuro/mental health	alcohol
U030	social	Do you use recreational drugs?	yes_no	affects mental health/neuro	drug_use
U031	context	Have you tried any medication or treatment for this problem already?	text	none	prior_treatment
U032	context	Have you had this problem before?	yes_no	none	recurrence
U033	travel	Have you traveled recently?	yes_no/text	triggers malaria/travel modifier	travel

6. MASTER RED-FLAG MODIFIER TABLE

These should be background rules developers apply across many symptom blocks.

Rule ID	Trigger	Action
RF001	severe breathing difficulty at any point	stop immediately
RF002	severe chest pain at any point	stop immediately
RF003	headache + worst headache of life	stop immediately
RF004	headache + confusion	stop immediately
RF005	headache + one-sided weakness / speech difficulty	stop immediately
RF006	sore throat + drooling	stop immediately
RF007	sore throat + unable to swallow liquids or saliva	stop immediately
RF008	eye symptoms + vision loss	stop immediately
RF009	eye symptoms + severe pain	stop immediately
RF010	vomiting + unable to keep fluids down + dizziness	stop immediately
RF011	diarrhea + severe dehydration	stop immediately
RF012	urinary symptoms + fever + flank pain + vomiting	urgent/in-person or stop depending severity
RF013	vaginal bleeding + heavy bleeding + dizziness	stop immediately
RF014	pelvic pain + pregnancy possible	urgent review / often stop
RF015	rash + facial swelling or breathing difficulty	stop immediately
RF016	hives + lip/tongue swelling	stop immediately
RF017	skin infection + rapidly spreading redness + fever	urgent same-day/in-person
RF018	back pain + loss of bowel/bladder control	stop immediately

RF019	leg pain + swelling + recent travel/immobility	urgent same-day review
RF020	depression/anxiety/mood changes + self-harm thoughts	stop immediately
RF021	weakness + one-sided symptoms	stop immediately
RF022	memory problem + acute confusion	stop immediately
RF023	penile/groin pain + severe testicular pain	stop immediately
RF024	black stool + dizziness/weakness	stop immediately
RF025	coughing blood	urgent review or stop depending severity

---

7. SYMPTOM BLOCKS — DATABASE-READY MASTER TABLE

Below is the **developer-ready symptom logic** based only on the symptoms in your document.

7.1 GENERAL SYMPTOMS

FEVER

ID	Question	Type	Ask If	Stop / Escalate
F001	When did the fever begin?	text/date	selected fever	none
F002	What is the highest temperature recorded?	number	fever	stop if extremely high + severe symptoms
F003	How did you measure the temperature?	select	fever	none
F004	Is the fever constant or on and off?	select	fever	none

F005	Do you have chills?	yes_no	fever	branch chills
F006	Do you have cough?	yes_no	fever	branch cough
F007	Do you have sore throat?	yes_no	fever	branch throat
F008	Do you have headache?	yes_no	fever	branch headache
F009	Do you have vomiting?	yes_no	fever	branch vomiting
F010	Do you have diarrhea?	yes_no	fever	branch diarrhea
F011	Do you have neck stiffness?	yes_no	fever	stop if yes with severe headache/confusion
F012	Do you feel confused or disoriented?	yes_no	fever	stop if yes
F013	Do you have severe trouble breathing?	yes_no	fever	stop if yes
F014	Do you have severe chest pain?	yes_no	fever	stop if yes
F015	Do you have immune system problems or take chemotherapy, transplant medication, or long-term steroids?	yes_no	fever	urgent same-day review if yes

CHILLS

ID	Question	Type	Ask If	Stop / Escalate
C001	When did the chills begin?	text	chills	none
C002	Do you also have fever?	yes_no	chills	branch fever
C003	Are the chills severe or shaking?	yes_no	chills	escalate if severe

C004	Do you have painful urination?	yes_no	chills	branch urinary
C005	Do you have cough?	yes_no	chills	branch cough
C006	Do you feel confused?	yes_no	chills	stop if yes

FATIGUE

ID	Question	Type	Stop / Escalate
FT001	When did the fatigue begin?	text	none
FT002	Did it start suddenly or gradually?	select	none
FT003	Is it constant or intermittent?	select	none
FT004	Are you sleeping poorly?	yes_no	branch insomnia
FT005	Do you have fever?	yes_no	branch fever
FT006	Have you lost weight?	yes_no	urgent if significant
FT007	Do you feel short of breath?	yes_no	stop if severe
FT008	Have you fainted?	yes_no	stop if yes
FT009	Are you too weak to stand or walk normally?	yes_no	stop if yes
FT010	Do you feel depressed?	yes_no	branch depression

WEAKNESS

ID	Question	Type	Stop / Escalate
W001	When did the weakness begin?	text	none

W002	Is the weakness one-sided or all over?	select	stop if one-sided sudden
W003	Does it affect your arms, legs, or both?	select	none
W004	Is it getting worse?	yes_no	escalate if worsening
W005	Do you have numbness?	yes_no	stop if sudden with one-sided weakness
W006	Do you have trouble speaking?	yes_no	stop if yes
W007	Did this start suddenly?	yes_no	stop if yes + focal weakness
W008	Are you able to walk normally?	yes_no	urgent if no

NIGHT SWEATS / WEIGHT LOSS / LOSS OF APPETITE / DEHYDRATION

Use the same structured pattern:

- onset
- severity/frequency
- associated fever
- cough
- GI symptoms
- travel
- ability to hydrate
- dizziness
- pregnancy if relevant
- urgent if severe dehydration, rapid major weight loss, night sweats with cough/fever/weight loss

---

7.2 HEAD SYMPTOMS

HEADACHE

ID	Question	Type	Stop / Escalate

H001	When did the headache begin?	text	none
H002	Was it sudden or gradual?	select	stop if sudden thunderclap
H003	Where is the pain located?	select/text	none
H004	Rate the pain from 1 to 10.	number	escalate if very severe
H005	Is this the worst headache of your life?	yes_no	stop if yes
H006	Do you have nausea or vomiting?	yes_no	none
H007	Are you sensitive to light?	yes_no	none
H008	Do you have fever?	yes_no	branch fever
H009	Do you have neck stiffness?	yes_no	stop if yes with fever
H010	Any blurred vision?	yes_no	urgent if severe
H011	Any one-sided weakness or numbness?	yes_no	stop if yes
H012	Any trouble speaking?	yes_no	stop if yes
H013	Any recent head injury?	yes_no	stop if significant/severe
H014	Have you had migraines before?	yes_no	none
H015	Are you confused or unusually sleepy?	yes_no	stop if yes

MEMORY PROBLEMS

Red flags here must be strong.

ID	Question	Type	Stop / Escalate
M001	When did the memory problem begin?	text	none
M002	Did it start suddenly?	yes_no	urgent if yes
M003	Are you forgetting recent events?	yes_no	none

M004	Are other people noticing this?	yes_no	escalate
M005	Are you confused right now?	yes_no	stop if yes
M006	Any recent head injury?	yes_no	urgent if yes
M007	Any trouble speaking?	yes_no	stop if yes
M008	Any weakness or numbness?	yes_no	stop if yes

7.3 EYE SYMPTOMS

EYE REDNESS / EYE DISCHARGE / ITCHY EYES

For all eye complaints, photo upload should be strongly encouraged, and required for redness/discharge if the platform policy allows.

ID	Question	Type	Stop / Escalate
E001	When did the eye problem begin?	text	none
E002	Is it one eye or both?	select	none
E003	Do you have discharge?	yes_no	none
E004	What color is the discharge?	select	none
E005	Is there itching?	yes_no	none
E006	Is there pain?	yes_no	stop if severe pain
E007	Is your vision blurry or changed?	yes_no	stop if major vision change

E008	Are you sensitive to light?	yes_no	urgent if yes + pain
E009	Do you wear contact lenses?	yes_no	urgent same-day if redness/pain
E010	Have you had trauma to the eye?	yes_no	urgent/stop if significant
E011	Please upload at least 2 clear photos of the affected eye.	photo	required if redness/discharge

---

7.4 EAR SYMPTOMS

EAR DISCHARGE

ID	Question	Type	Stop / Escalate
ER001	When did the discharge begin?	text	none
ER002	What color is the discharge?	select	none
ER003	Do you have ear pain?	yes_no	none
ER004	Do you have fever?	yes_no	escalate if severe
ER005	Have you noticed hearing loss?	yes_no	urgent if severe/sudden
ER006	Any recent trauma or injury?	yes_no	urgent if yes
ER007	Any swelling behind the ear?	yes_no	urgent same-day

7.5 NOSE / THROAT / CHEST

RUNNY NOSE / NASAL CONGESTION / SINUS PRESSURE / SNEEZING

Questions should merge efficiently.

Core questions:

- onset
- clear vs colored discharge
- sneezing
- itchy eyes
- facial pain/pressure
- fever
- cough
- tooth pain
- swelling around eyes
- severe headache

Key red flags:

- facial swelling
- eye swelling with pain/vision change
- severe headache with fever and confusion
- severe breathing issue

SORE THROAT / HOARSENESS / DIFFICULTY SWALLOWING / THROAT IRRITATION

ID	Question	Type	Stop / Escalate
T001	When did the sore throat or throat problem begin?	text	none
T002	Rate the throat pain 1 to 10.	number	none
T003	Do you have fever?	yes_no	none
T004	Do you have cough?	yes_no	none
T005	Do you have runny nose?	yes_no	none
T006	Is it painful to swallow?	yes_no	none
T007	Are you able to swallow liquids?	yes_no	stop if no
T008	Are you drooling or unable to swallow saliva?	yes_no	stop if yes
T009	Do you have trouble breathing?	yes_no	stop if yes

T010	Do you see white patches on your throat?	yes_no	none
T011	Please upload 2 clear throat photos if prompted.	photo	required for strep-like throat complaints

COUGH / PERSISTENT COUGH / COUGH WITH MUCUS

ID	Question	Type	Stop / Escalate
CO001	When did the cough start?	text	none
CO002	Is it dry or with mucus?	select	none
CO003	What color is the mucus?	select	none
CO004	Do you have fever?	yes_no	none
CO005	Do you have shortness of breath?	yes_no	stop if severe
CO006	Do you have chest pain?	yes_no	stop if severe
CO007	Are you coughing up blood?	yes_no	stop or urgent same-day depending amount
CO008	Do you smoke?	yes_no	none
CO009	Do you have asthma?	yes_no	urgent if worsening breathing
CO010	How long have you had the cough?	text	urgent if prolonged + weight loss/night sweats
CO011	Have you lost weight?	yes_no	urgent if yes with prolonged cough
CO012	Do you have night sweats?	yes_no	urgent if yes with prolonged cough

CO013	Are you wheezing?	yes_no	urgent if worsening

7.6 GASTROINTESTINAL

NAUSEA / VOMITING / DIARRHEA / CONSTIPATION / BLOATING / EXCESS GAS / HEARTBURN / INDIGESTION

Strong red flags are crucial here.

VOMITING

ID	Question	Type	Stop / Escalate
V001	When did vomiting begin?	text	none
V002	How many times have you vomited today?	number	escalate if frequent
V003	Can you keep fluids down?	yes_no	stop if no
V004	Do you feel dizzy or faint?	yes_no	stop if severe
V005	Is there blood in the vomit?	yes_no	stop if yes
V006	Do you have severe abdominal pain?	yes_no	stop if yes
V007	Could you be pregnant?	yes_no	urgent if yes with persistent vomiting
V008	Do you have diarrhea?	yes_no	none

DIARRHEA

ID	Question	Type	Stop / Escalate
D001	When did diarrhea begin?	text	none
D002	How many bowel movements per day?	number	none
D003	Is there blood in the stool?	yes_no	urgent / stop if significant

D004	Is the stool black?	yes_no	stop if yes with weakness/dizziness
D005	Do you have fever?	yes_no	none
D006	Do you have severe abdominal pain?	yes_no	stop if yes
D007	Can you drink fluids?	yes_no	stop if no
D008	Have you traveled recently?	yes_no	none
D009	Have you taken antibiotics recently?	yes_no	urgent if severe diarrhea

CONSTIPATION

Red flags:

- inability to pass gas
- vomiting
- severe abdominal distention/pain
- blood in stool with weakness

HEARTBURN / INDIGESTION

Red flags:

- difficulty swallowing
- vomiting blood
- black stools
- severe chest pain
- severe abdominal pain

7.7 URINARY

PAINFUL URINATION / FREQUENT URINATION / URGENT URINATION / NIGHTTIME URINATION

ID	Question	Type	Stop / Escalate
UTI001	When did urinary symptoms begin?	text	none
UTI002	Is there burning with urination?	yes_no	none

UTI003	Are you urinating more often?	yes_no	none
UTI004	Do you have urgency?	yes_no	none
UTI005	Is there blood in the urine?	yes_no	urgent if significant
UTI006	Do you have fever?	yes_no	urgent if yes
UTI007	Do you have pain in your back or side near the kidneys?	yes_no	urgent if yes
UTI008	Are you vomiting?	yes_no	urgent/stop if with fever + flank pain
UTI009	Could you be pregnant?	yes_no	urgent same-day if yes
UTI010	Have you had UTIs before?	yes_no	none

Rule:

- fever + flank pain + vomiting = not routine asynchronous, urgent or stop
- pregnancy + urinary symptoms = physician priority escalation

---

7.8 FEMALE REPRODUCTIVE

VAGINAL DISCHARGE / ITCHING / ODOR / BLEEDING / PELVIC PAIN / PAIN DURING INTERCOURSE / IRREGULAR PERIODS

ID	Question	Type	Stop / Escalate
GYN001	When did the symptom begin?	text	none
GYN002	What color is the discharge?	select	none
GYN003	Is there an odor?	yes_no	none
GYN004	Is there itching?	yes_no	none

GYN005	Do you have pelvic pain?	yes_no	urgent if severe
GYN006	Do you have fever?	yes_no	urgent same-day
GYN007	Could you be pregnant?	yes_no	urgent if bleeding/pelvic pain
GYN008	Are you bleeding heavily?	yes_no	stop if yes with dizziness/heavy pad soaking
GYN009	Are you feeling dizzy or faint?	yes_no	stop if yes with bleeding
GYN010	Do you have pain during sex?	yes_no	none
GYN011	Any new sexual partner?	yes_no	none
GYN012	When was your last menstrual period?	date/text	none

Critical rules:

- pregnancy possible + pelvic pain = urgent evaluation
- pregnancy possible + bleeding = urgent evaluation
- heavy bleeding + dizziness = stop immediately
- pelvic pain + fever = urgent same-day

7.9 MALE REPRODUCTIVE

GROIN PAIN / PENILE DISCHARGE / ERECTILE PROBLEMS / PAIN DURING EJACULATION

ID	Question	Type	Stop / Escalate
MEN001	When did the symptom begin?	text	none
MEN002	Is the pain on one side or both?	select	none
MEN003	Is there testicular pain?	yes_no	stop if severe sudden
MEN004	Is there swelling?	yes_no	urgent if yes
MEN005	Is there penile discharge?	yes_no	none

MEN006	Any burning with urination?	yes_no	none
MEN007	Any fever?	yes_no	urgent if yes
MEN008	Any recent sexual contact or STI exposure?	yes_no	none

Critical rule:

- sudden severe unilateral testicular pain = emergency stop

7.10 SKIN

RASH / SKIN REDNESS / SKIN INFECTION / SKIN ITCHING / BLISTERS / SKIN BUMPS / HAIR LOSS / HIVES / SWELLING

All visible skin complaints should support required image upload where appropriate.

ID	Question	Type	Stop / Escalate
SK001	When did the skin problem begin?	text	none
SK002	Where is it located?	text/select	none
SK003	Is it itchy?	yes_no	none
SK004	Is it painful?	yes_no	none
SK005	Is it spreading?	yes_no	urgent if rapid
SK006	Do you have fever?	yes_no	urgent if yes with rash/infection
SK007	Is there pus or drainage?	yes_no	urgent if yes

SK008	Is there warmth or swelling?	yes_no	urgent if yes
SK009	Have you started a new medication?	yes_no	urgent if rash after new medication
SK010	Do you have swelling of your face, lips, or tongue?	yes_no	stop if yes
SK011	Do you have trouble breathing?	yes_no	stop if yes
SK012	Please upload at least 2 to 3 clear photos.	photo	required for rash/infection/blisters/bumps

Key red flags:

- rash + trouble breathing
- hives + lip/tongue swelling
- rapidly spreading redness + fever
- severe painful blistering rash after medication
- facial rash/swelling with airway symptoms
- infection near eye with swelling

7.11 MUSCULOSKELETAL

JOINT PAIN / JOINT SWELLING / MUSCLE PAIN / MUSCLE WEAKNESS / BACK PAIN / NECK PAIN / SHOULDER / KNEE / HIP / FOOT / ARM / LEG PAIN / LIMITED MOBILITY

ID	Question	Type	Stop / Escalate
MSK001	Which body part is affected?	select	none
MSK002	When did symptoms begin?	text	none
MSK003	Was there an injury?	yes_no	none
MSK004	Is there swelling?	yes_no	none
MSK005	Is there redness or warmth?	yes_no	urgent if yes with fever

MSK006	Do you have fever?	yes_no	urgent if yes
MSK007	Can you move the joint normally?	yes_no	urgent if unable
MSK008	Can you bear weight?	yes_no	urgent if unable
MSK009	Any numbness or tingling?	yes_no	urgent/stop if severe neuro signs
MSK010	Any weakness?	yes_no	stop if one-sided sudden
MSK011	Any loss of bowel or bladder control?	yes_no	stop if yes
MSK012	Any recent long travel or immobilization?	yes_no	urgent if leg pain/swelling

Key rules:

- back pain + bowel/bladder loss = stop immediately
- neck pain + fever + severe headache = stop
- leg pain + swelling + travel/immobility = urgent same-day
- joint swelling + fever = urgent same-day
- inability to bear weight after injury = urgent in-person

7.12 MENTAL HEALTH

ANXIETY / DEPRESSION / MOOD CHANGES / DIFFICULTY SLEEPING

ID	Question	Type	Stop / Escalate
MH001	When did the symptoms begin?	text	none
MH002	Is it constant or does it come in episodes?	select	none
MH003	Are you having panic attacks?	yes_no	none
MH004	Are you sleeping poorly?	yes_no	none

MH005	Have you lost interest in usual activities?	yes_no	none
MH006	Do you feel hopeless?	yes_no	urgent if severe
MH007	Are you having thoughts of harming yourself?	yes_no	stop if yes
MH008	Are you having thoughts of harming someone else?	yes_no	stop if yes
MH009	Are you hearing or seeing things others do not?	yes_no	urgent/stop depending severity
MH010	Are you unable to function safely at home or work?	yes_no	urgent same-day

---

8. CONDITION BLOCKS — MASTER FORMAT

Now the same style for condition-based entry.

The platform should still ask structured safety questions even if a patient selects a condition.

8.1 UPPER RESPIRATORY / FLU / COVID / BRONCHITIS / ALLERGIC RHINITIS / SINUSITIS

Core questions:

- onset
- fever
- cough
- mucus
- wheezing
- shortness of breath
- facial pain/pressure
- sore throat
- recent sick contacts

- smoking/asthma history

Critical red flags:

- breathing difficulty
- chest pain
- confusion
- severe dehydration
- coughing blood
- facial swelling with severe pain

8.2 SORE THROAT / PHARYNGITIS / LARYNGITIS / POST NASAL DRIP

Add:

- swallowing ability
- drooling
- fever
- white patches
- cough/runny nose
- photo upload if throat infection suspected

Critical red flags:

- inability to swallow liquids
- drooling
- breathing difficulty
- muffled voice with severe throat pain

8.3 CONJUNCTIVITIS / EYE DRYNESS / STYE

Add:

- contact lens use
- eye pain
- vision change
- light sensitivity
- photo upload

Critical red flags:

- severe eye pain
- vision loss
- trauma
- contact lens wearer with redness and pain

8.4 GASTROENTERITIS / FOOD POISONING / GERD / GASTRITIS / CONSTIPATION / DIARRHEA / BLOATING / INDIGESTION

Critical red flags:

- vomiting blood
- black stool
- severe abdominal pain
- inability to keep fluids down
- dehydration with dizziness
- pregnancy with severe vomiting
- blood in stool with weakness

8.5 UTI / BLADDER INFECTION

Critical red flags:

- fever
- flank pain
- vomiting
- pregnancy
- inability to urinate
- severe illness appearance

8.6 CHLAMYDIA / GONORRHEA

Required attestation must be enforced before questions.

Mandatory attestation:

I confirm that I either tested positive for this infection or had contact with someone who tested positive.

Questions:

- test positive or exposed
- symptoms present
- discharge
- burning
- pelvic/groin pain
- pregnancy
- fever

Critical red flags:

- pelvic pain + fever
- testicular pain/swelling
- pregnancy with STI symptoms/exposure
- severe abdominal pain

8.7 YEAST INFECTION / BV / VAGINAL DISCHARGE / PAINFUL MENSTRUATION / MENOPAUSE / VAGINAL DRYNESS

Critical red flags:

- severe pelvic pain
- heavy bleeding
- fever
- pregnancy possible
- fainting/dizziness with bleeding

8.8 BIRTH CONTROL / EMERGENCY CONTRACEPTION

Birth control consult must require negative pregnancy test confirmation.

Mandatory attestation:

I confirm that my pregnancy test result is negative.

Questions:

- pregnancy test date
- result
- smoking
- migraine with aura history
- blood clot history
- high blood pressure history

Critical red flags / exclusions:

- possible pregnancy not excluded
- heavy unexplained bleeding
- severe migraine with neurologic symptoms
- active clot concerns
- severe chest pain / shortness of breath

8.9 ED / PREMATURE EJACULATION

Questions:

- onset
- constant vs intermittent
- medication history
- diabetes/HTN history
- chest pain with sex?
- current nitrate use?

Critical issues:

- chest pain symptoms
- nitrate use if certain meds considered
- penile deformity/painful erection if later medication library supports ED treatment

---

8.10 DERMATOLOGY CONDITIONS

Acne, rosacea, eczema, psoriasis, hives, cellulitis, boils, folliculitis, fungal infections, athlete's foot, ringworm, nail fungus, warts, cold sores, early shingles, burns, seborrheic dermatitis, dandruff, hair loss, eyelash loss.

Critical red flags:

- facial swelling
- airway symptoms
- rapidly spreading cellulitis
- fever with skin infection
- rash after new medication with systemic symptoms
- shingles near eye
- burn that is extensive or severe
- painful facial infection near eye

All visible lesions should support photo upload.

8.11 MUSCULOSKELETAL CONDITIONS

Acute back pain, neck pain, muscle strain/spasm, shoulder/knee/hip/ankle/wrist pain, sciatica, sprains.

Critical red flags:

- bowel/bladder loss
- saddle anesthesia if platform later adds it
- one-sided weakness
- inability to bear weight
- severe trauma
- fever with spinal pain
- major swelling/deformity

8.12 MIGRAINE

Questions:

- similar to prior migraines?
- aura?
- nausea
- light sensitivity
- neuro symptoms
- worst headache of life

Critical red flags:

- first/worst severe headache
- confusion
- weakness
- speech issues
- fever + stiff neck
- head trauma

8.13 ANXIETY / DEPRESSION / INSOMNIA

Critical red flags:

- suicidality
- homicidality
- psychosis
- inability to care for self safely

8.14 GENERAL SERVICES

Medication refill, medication side effects, lab result review, imaging review, second opinion, smoking cessation, malaria prophylaxis, sea sickness prevention, EpiPen.

Special safety rules:

- **med refill**: no controlled substances
- **med side effects**: stop if severe allergic reaction, fainting, chest pain, facial swelling, severe rash
- **malaria prophylaxis**: travel modifier questions, medication contraindications
- **EpiPen**: not for current anaphylaxis; if currently symptomatic stop immediately

---

9. PHOTO ENFORCEMENT RULES

Mandatory photos for:

- rash
- skin redness
- skin infection
- blisters

- skin bumps
- acne
- cold sores
- shingles
- eye redness
- eye discharge
- sore throat when requested by protocol
- burns

Photo rules:

Rule	Requirement
P001	minimum 2 photos
P002	one close-up
P003	one wider/context image
P004	good lighting required
P005	block submission if required photos missing

---

10. FINAL SAFETY SCAN BEFORE SUBMISSION

Before the patient submits, run a final background rules engine.

Examples:

- headache + “worst of life” = stop
- vomiting + cannot keep fluids down = stop
- vaginal bleeding + dizziness = stop
- sore throat + cannot swallow liquids = stop
- rash + facial swelling = stop
- urinary symptoms + fever + flank pain + vomiting = urgent/in-person
- depression + self-harm thoughts = stop

- eye pain + vision change = stop
- leg swelling + pain + immobility = urgent same-day

If triggered, patient should not proceed to normal submission.

---

11. AI INTERNAL SUMMARY MAPPING

Each answer should map into structured physician fields:

- chief complaint
- duration
- severity
- associated symptoms
- relevant positives
- relevant negatives
- red flags checked
- red flags denied
- medical history
- pregnancy status
- allergies
- medications
- photos attached
- AI suggested differential
- AI suggested treatment options
- AI suggested tests
- disposition:
    - routine async
    - urgent same-day physician review
    - in-person care advised
    - emergency stop

---

12. AI SOAP NOTE GENERATOR TEMPLATE

Subjective

- age/sex
- chief complaint
- duration
- symptom details
- relevant positives
- relevant negatives
- PMH
- meds
- allergies
- pregnancy status if relevant

Objective

- asynchronous consult
- no in-person physical exam
- patient-reported findings
- uploaded photo review if present

Assessment

- AI suggestions for physician only
- doctor edits final assessment

Plan

- treatment options
- testing suggestions
- prescription if applicable
- precautions
- follow-up instructions
- escalation instructions

---

13. PHYSICIAN DASHBOARD — COMPLETE FLOW

Main Queues

- new
- urgent
- awaiting clarification
- ready for prescribing
- completed
- escalated

Case View Sections

- patient header
- symptom/condition selected
- AI summary
- answers timeline
- photo gallery
- red flag review
- AI suggestions
- physician note editor
- SOAP draft
- prescription panel
- patient message panel

Quick Buttons

- approve
- edit
- request more info
- prescribe
- send advice only
- recommend urgent care
- send ER message
- close case

---

14. AUTOMATED PHARMACY PRESCRIPTION WORKFLOW

This must remain physician-controlled.

Required Pharmacy Fields

- pharmacy name
- address
- city/state
- phone
- eRx endpoint if integrated
- backup pharmacy

Prescription Guardrails

Block or flag:

- opioids
- benzodiazepines
- ADHD stimulants
- controlled sleep meds
- restricted muscle relaxants
- lab-monitoring meds outside platform rules

Safety Checks

- allergy cross-check
- pregnancy cross-check
- age restrictions
- duplicate therapy check
- contraindication check
- interaction check where supported

---

15. EXAMPLE JSON RULE OBJECT


```
{
  "question_id": "H005",
  "module_type": "symptom",
  "parent_topic": "headache",
  "question_text": "Is this the worst headache of your life?",
  "answer_type": "yes_no",
  "required": true,
  "ask_order": 5,
  "ask_if": ["selected_symptom=headache"],
  "stop_if": ["answer=yes"],
  "escalate_if": [],
```

"summary_map": "headache_worst_of_life",
"negative_map": "denies_worst_headache",
"photo_required_if": [],
"notes_for_devs": "Hard stop because possible hemorrhage or other dangerous cause."
}

---

16. EXAMPLE TRIAGE OUTPUT JSON

{
"case_id": "12345",
"entry_mode": "symptom",
"selected_symptoms": ["painful urination", "frequent urination", "urgent urination"],
"priority": "moderate",
"disposition": "routine_async",
"red_flags_checked": [
"severe chest pain denied",
"difficulty breathing denied",
"stroke symptoms denied",
"severe abdominal pain denied"
],
"red_flags_triggered": [],
"history": {
"duration": "2 days",
"severity": "moderate",
"associated_symptoms": ["frequency", "urgency"],
"relevant_negatives": ["no fever", "no flank pain", "no vomiting"]
},
"history_medical": {
"pregnancy_status": "not pregnant",
"allergies": ["none reported"],
"current_medications": []
},
"ai_suggestions": {
"possible_differentials": ["acute cystitis", "urethritis", "vaginitis"],
"possible_treatments": ["nitrofurantoin", "TMP-SMX", "phenazopyridine"],
"possible_tests": ["urinalysis if needed"]
}
}

---

17. HOW TO SCALE TO 1,200+ QUESTIONS CORRECTLY

You do **not** need 1,200 totally unrelated questions.
You need **1,200+ reusable structured question objects**.

That total will come from:

- all symptom questions
- all condition questions
- all modifier questions
- all safety questions
- all pregnancy/travel/STI/refill/photo questions
- all prescribing safety questions
- all follow-up clarification questions

A safe build pattern is:

Tier 1

Universal + symptom core bank

Tier 2

Condition-specific logic

Tier 3

Modifier blocks

Tier 4

Prescription guardrail questions

Tier 5

Clarification follow-up bank

---

18. WHAT YOUR DEVELOPERS SHOULD BUILD NEXT

The most useful next artifact is a **true import-ready master spreadsheet/CSV schema** with columns like:

- question_id
- module_type
- parent_topic
- entry_type
- question_text
- answer_type
- answer_options
- required
- ask_order
- ask_if
- stop_if
- escalate_if
- summary_map
- negative_map
- photo_required_if
- physician_only_flag
- patient_visible_flag
- active_status